Introduction to Matrix Calculus

Slide edition

Expanded "A Gentle Introduction to Matrix Calculus"
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1 The tools

Six tools for matrix calculus.

1.1 The trace

For square $A = (a_{ij})$, $\operatorname{tr} A = \sum_i a_{ii}$. Immediately,

$$\operatorname{tr} A' = \operatorname{tr} A. \tag{1}$$

Proposition 1. *For any two matrices A and B of the same order,*

$$\operatorname{tr} A'B = \operatorname{tr} BA'.$$

Proof. This follows because

$$\operatorname{tr} A'B = \sum_j (A'B)_{jj} = \sum_j \sum_i a_{ij} b_{ij} = \sum_i \sum_j b_{ij} a_{ij} = \sum_i (BA')_{ii} = \operatorname{tr} BA'.$$

□

Hence cyclical: $\operatorname{tr} ABC = \operatorname{tr} CAB = \operatorname{tr} BCA$ (but $\operatorname{tr} ABC \neq \operatorname{tr} ACB$ in general).

1.2 Linear and quadratic forms

Linear form Ax. $Ax = 0$ does not force $A = 0$ or $x = 0$. But $Ax = 0$ for *every* $x \iff A = 0$.

Quadratic form $x'Ax$. Even $x'Ax = 0$ for every x does not force $A = 0$: the skew-symmetric $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is a

counterexample. In general, $x'Ax = 0$ for every $x \iff A$ is skew-symmetric ($A' = -A$).

Proposition 2. *We have*
(i) $Ax = 0$ *for every* $x \iff A = 0$,
(ii) $x'Ax = 0$ *for every* $x \iff A$ *is skew-symmetric,*
(iii) $x'Ax = 0$ *for every* x *and* $A = A' \iff A = 0$.

Proof. We shall prove both directions of these equivalences to ensure completeness. Let e_i denote the ith elementary vector, that is, the vector with one in the ith position and zeros elsewhere.
Proof of (i). First assume $A = 0$. Then $Ax = 0x = 0$ for all x. Conversely, assume $Ax = 0$ for every x. Let $x = e_i$. Then $Ae_i = 0$ for every i, meaning every column of A is zero, and hence $A = 0$.
Proof of (ii). First assume A is skew-symmetric, so $A = -A'$. Since $x'Ax$ is a scalar, it equals its own transpose: $x'Ax = (x'Ax)' = x'A'x$. Substituting $A' = -A$ gives $x'Ax = -x'Ax$, which implies $2x'Ax = 0$, so $x'Ax = 0$ for all x. Conversely, assume $x'Ax = 0$ for every x. Setting $x = e_i$ directly gives $e_i'Ae_i = a_{ii} = 0$ for every i, so every diagonal entry vanishes. Next, setting $x = e_i + e_j$ for $i \neq j$ yields

$$0 = (e_i + e_j)'A(e_i + e_j) = a_{ii} + a_{ij} + a_{ji} + a_{jj}.$$

Since $a_{ii} = a_{jj} = 0$, we get $a_{ij} + a_{ji} = 0$. Hence $A + A' = 0$, i.e., A is skew-symmetric.
Proof of (iii). First assume $A = 0$. Then $A = A'$ and $x'Ax = 0$ for every x. Conversely, assume $x'Ax = 0$ for every x and $A = A'$. From (ii), $x'Ax = 0$ for every x implies $A = -A'$. Since we also have $A = A'$, it follows that $A = -A$, which implies $2A = 0$, so A must be the null matrix. □

Remark 1 (Use).
- (i): to show $A = B$, it suffices to show $Ax = Bx$ for all x.
- (ii): $x'Ax = x'Bx$ for all $x \implies A + A' = B + B'$ (not $A = B$); with symmetry, $A = (B + B')/2$. Used in Prop. 16.

1.3 The Kronecker product

For A $(m \times n)$ and B $(p \times q)$,

$$A \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \dots & a_{mn}B \end{pmatrix} \qquad (mp \times nq). \tag{2}$$

Defined for *any* pair A, B (unlike AB).

Proposition 3. *The following three properties show that the Kronecker product is in fact a product:*

$$\begin{aligned} (A \otimes B) \otimes C &= A \otimes (B \otimes C), \\ (A + B) \otimes (C + D) &= A \otimes C + A \otimes D + B \otimes C + B \otimes D, \\ (A \otimes B)(C \otimes D) &= AC \otimes BD, \end{aligned}$$

where it is assumed that $A + B$ and $C + D$ are defined in the second equality, and that AC and BD are defined in the third equality.

Proof. We prove each equality in turn, using the fact that the (i, j) block of $A \otimes B$ is $a_{ij}B$.
Associativity. Let A, B, C have orders $m \times n$, $p \times q$, $r \times s$, respectively. Both sides are $mpr \times nqs$ matrices. The (i, j) macro-block of $(A \otimes B) \otimes C$ is $(a_{ij}B) \otimes C$. Since a_{ij} is a scalar, we can factor it out to yield $a_{ij}(B \otimes C)$. This expression is precisely the (i, j) block of $A \otimes (B \otimes C)$. Because all corresponding blocks are identical, the two matrices are equal.
Distributivity. The (i, j) block of $(A + B) \otimes (C + D)$ is

$$(a_{ij} + b_{ij})(C + D) = a_{ij}C + a_{ij}D + b_{ij}C + b_{ij}D,$$

which is the sum of the (i, j) blocks of $A \otimes C$, $A \otimes D$, $B \otimes C$, and $B \otimes D$.
Mixed-product property. Let A be $m \times n$, B be $p \times q$, C be $n \times s$, and D be $q \times t$. Both sides are $mp \times st$ matrices. Writing the product $(A \otimes B)(C \otimes D)$ in block form, the (i, k) block (of size $p \times t$) is

$$\sum_{j=1}^{n} (a_{ij}B)(c_{jk}D) = \Big(\sum_{j=1}^{n} a_{ij}c_{jk}\Big)BD = (AC)_{ik}\, BD,$$

which is exactly the (i, k) block of $AC \otimes BD$. □

Exercise 1. For any two column vectors a and b (not necessarily of the same order), we have

$$a \otimes b' = ab' = b' \otimes a.$$

Solution. This follows, in essence, from the fact that a vector x can be written as $x \otimes 1$ and also as $1 \otimes x$. Define elementary vectors e_i $(m \times 1)$ and u_j $(n \times 1)$, where m and n are the orders of a and b, respectively. Then,

$$e_i'(a \otimes b')u_j = (e_i' \otimes 1)(a \otimes b')(1 \otimes u_j) = (e_i'a) \otimes (b'u_j) = a_i b_j = (ab')_{ij},$$
$$e_i'(b' \otimes a)u_j = (1 \otimes e_i')(b' \otimes a)(u_j \otimes 1) = (b'u_j) \otimes (e_i'a) = b_j a_i = (ab')_{ij},$$

using Proposition 3. □

Proposition 4. *We have*

$$(A \otimes B)' = A' \otimes B',$$
$$\operatorname{tr}(A \otimes B) = (\operatorname{tr} A)(\operatorname{tr} B),$$
$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1},$$

where A and B are assumed to be square in the second equality and nonsingular in the third equality.

Proof. Transpose. The (i, j) block of $A \otimes B$ is $a_{ij}B$. The transpose exchanges blocks: the (j, i) block of $(A \otimes B)'$ is $(a_{ij}B)' = a_{ij}B'$. But the (j, i) block of $A' \otimes B'$ is $(A')_{ji}B' = a_{ij}B'$. Since all blocks agree, $(A \otimes B)' = A' \otimes B'$. *Trace.* The ith diagonal block of $A \otimes B$ is $a_{ii}B$ (a $p \times p$ matrix). The trace of $A \otimes B$ is the sum of the traces

of its diagonal blocks:

$$\operatorname{tr}(A \otimes B) = \sum_{i=1}^{m} \operatorname{tr}(a_{ii} B) = \sum_{i=1}^{m} a_{ii} \operatorname{tr} B = (\operatorname{tr} A)(\operatorname{tr} B).$$

Inverse. This follows from Proposition 3 by writing

$$(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I_m \otimes I_n = I_{mn},$$

where A has order $m \times m$, and B has order $n \times n$. □

1.4 The vec operator

For A $(m \times n)$ with columns $a_1, \ldots, a_n$, stack columns:

$$\operatorname{vec} A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \qquad (mn \times 1). \tag{3}$$

Exercise 2. If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix},$$

then what is $\operatorname{vec} A$? And what is $\operatorname{vec} A'$?

Solution. We have $\operatorname{vec} A = (1, 4, 2, 5, 3, 6)'$ and $\operatorname{vec} A' = (1, 2, 3, 4, 5, 6)'$. □

Proposition 5. *If A and B are matrices of the same order, then*

$$(\operatorname{vec} A)'(\operatorname{vec} B) = \operatorname{tr} A'B.$$

Proof. By definition, $\operatorname{vec} A$ and $\operatorname{vec} B$ are obtained by stacking the columns of A and B, so the inner product

$$(\operatorname{vec} A)'(\operatorname{vec} B) = \sum_{i,j} a_{ij} b_{ij}$$

multiplies corresponding elements and adds them up. From the proof of Proposition 1, the same double sum equals $\operatorname{tr} A'B$. Hence $(\operatorname{vec} A)'(\operatorname{vec} B) = \operatorname{tr} A'B$. □

Exercise 3. If a and b are vectors of arbitrary order, show that

$$\operatorname{vec} ab' = b \otimes a.$$

Solution. The jth column of ab' is $b_j a$. Stacking these columns, we get

$$\operatorname{vec}(ab') = \begin{pmatrix} b_1 a \\ b_2 a \\ \vdots \\ b_n a \end{pmatrix} = b \otimes a,$$

which matches the definition of the Kronecker product $b \otimes a$. □

Generalizing Exercise 3:

Proposition 6. *For any matrices A, B, and C for which the product ABC is defined, we have*

$$\operatorname{vec} ABC = (C' \otimes A) \operatorname{vec} B.$$

Proof. Using Exercise 3, we see that

$$\operatorname{vec}(Abe'C) = \operatorname{vec}\big((Ab)(C'e)'\big) = (C'e) \otimes (Ab) = (C' \otimes A)(e \otimes b) = (C' \otimes A) \operatorname{vec}(be')$$

for any vectors b and e. Then, writing $B = \sum_j b_j e_j'$ where b_j and e_j denote the jth column of B and I, respectively, the result follows. □

1.5 The commutation matrix

$\operatorname{vec} A$ and $\operatorname{vec} A'$ contain the same mn entries reordered. The unique $mn \times mn$ permutation K_{mn} satisfies

$$K_{mn} \operatorname{vec} A = \operatorname{vec} A'. \tag{4}$$

Write $K_n := K_{nn}$.

Exercise 4. Show that $\operatorname{tr} K_n = n$.

Solution. The ith diagonal entry of A occupies the same position in $\operatorname{vec} A$ and $\operatorname{vec} A'$, so each of the n diagonal blocks of K_n contributes a single 1. □

Proposition 7. *The commutation matrix satisfies*

$$K'_{mn} = K_{mn}^{-1} = K_{nm}.$$

Proof. Permutation matrices are orthogonal, so $K'_{mn} = K_{mn}^{-1}$. Premultiplying (4) by K_{nm} gives $K_{nm}K_{mn} \operatorname{vec} A = \operatorname{vec} A$, hence $K_{nm}K_{mn} = I_{mn}$. □

K lets us commute the factors of a Kronecker product:

Proposition 8.

$$K_{pm}(A \otimes B) = (B \otimes A)K_{qn}$$

for any $m \times n$ matrix A and $p \times q$ matrix B.

Proof. Test both sides against arbitrary $q \times n$ matrix X (Prop. 2(i)). Using (4) and Prop. 6,

$$K_{pm}(A \otimes B) \operatorname{vec} X = K_{pm} \operatorname{vec} BXA' = \operatorname{vec} AX'B' = (B \otimes A) \operatorname{vec} X' = (B \otimes A) K_{qn} \operatorname{vec} X. \qquad \square$$

Exercise 5. Let $N_n = (I_{n^2} + K_n)/2$. Show that N_n is symmetric idempotent and that $N_n(A \otimes A) = (A \otimes A)N_n$ for every $n \times n$ matrix A.

Solution. Since K_n is symmetric and orthogonal by Proposition 7, we have $K_n^2 = I_{n^2}$ and hence $N_n = N_n' = N_n^2$. Also, using Proposition 8, $K_n(A \otimes A) = (A \otimes A)K_n$ and hence $N_n(A \otimes A) = (A \otimes A)N_n$. $\square$

1.6 The duplication matrix

$\operatorname{vech}(A)$ stacks A's on-and-below-diagonal entries, e.g. for $n = 3$:

$$\begin{pmatrix} a_{11} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & a_{22} & \cancel{a_{23}} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \implies \operatorname{vech}(A) = (a_{11}, a_{21}, a_{31}, a_{22}, a_{32}, a_{33})'.$$

For symmetric A, $\operatorname{vec} A$ is $\operatorname{vech}(A)$ with repeats. The duplication matrix D_n $\left(n^2 \times \frac{1}{2}n(n+1)\right)$ satisfies

$$D_n \operatorname{vech}(A) = \operatorname{vec} A \qquad (A = A'). \tag{5}$$

D_n has full column rank, so $D_n' D_n$ is invertible and

$$\operatorname{vech}(A) = (D_n' D_n)^{-1} D_n' \operatorname{vec} A \qquad (A = A'). \tag{6}$$

Proposition 9. *The duplication matrix is connected to the commutation matrix by*

$$K_n D_n = D_n, \qquad D_n(D_n' D_n)^{-1} D_n' = \frac{1}{2}(I_{n^2} + K_n).$$

Proof. Let X be a symmetric $n \times n$ matrix. Then,

$$K_n D_n \operatorname{vech}(X) = K_n \operatorname{vec} X = \operatorname{vec} X = D_n \operatorname{vech}(X).$$

The symmetry of X does not restrict $\operatorname{vech}(X)$, which is therefore arbitrary. Hence, the first result follows. To prove the second result, let

$$M_n = D_n(D_n' D_n)^{-1} D_n', \qquad N_n = \tfrac{1}{2}(I_{n^2} + K_n), \qquad \Delta_n = M_n - N_n.$$

Both M_n and N_n are symmetric idempotent, the latter by Exercise 5. Since $K_n D_n = D_n$, we have $M_n N_n = N_n M_n = M_n$, so that Δ_n is also symmetric idempotent. Now,

$$\operatorname{tr} M_n = \operatorname{tr} I_{n(n+1)/2} = n(n+1)/2 = (n^2 + n)/2 = \operatorname{tr} N_n,$$

since $\operatorname{tr} K_n = n$ (Exercise 4). Since Δ_n is symmetric idempotent, $\mathrm{r}(\Delta_n) = \operatorname{tr} \Delta_n = 0$, and hence $\Delta_n = 0$. □

The matrix $D_n'(A \otimes A) D_n$ (cf. Exercise 28, Section 12):

Proposition 10. *Let A be an $n \times n$ matrix, not necessarily symmetric. The matrix $D_n'(A \otimes A)D_n$ satisfies the following properties:*

$$\begin{aligned} D_n(D_n'D_n)^{-1}D_n'(A \otimes A)D_n &= (A \otimes A)D_n, \\ (D_n'(A \otimes A)D_n)^{-1} &= (D_n'D_n)^{-1}D_n'(A^{-1} \otimes A^{-1})D_n(D_n'D_n)^{-1}, \\ |D_n'(A \otimes A)D_n| &= 2^{\frac{1}{2}n(n-1)}|A|^{n+1}, \end{aligned}$$

where A is assumed to be nonsingular in the second equality.

Proof. The first result follows from Exercise 5 and $N_nD_n = D_n$. To prove the second result we write

$$\begin{aligned} &D_n'(A \otimes A)D_n(D_n'D_n)^{-1}D_n'(A^{-1} \otimes A^{-1})D_n(D_n'D_n)^{-1} \\ &\quad = D_n'(A \otimes A)(A^{-1} \otimes A^{-1})D_n(D_n'D_n)^{-1} \\ &\quad = D_n'D_n(D_n'D_n)^{-1} = I_{n(n+1)/2}. \end{aligned}$$

To prove the third result, we use the multiplicative property of the determinant:

$$|D_n'(A \otimes A)D_n| = |D_n'D_n|\,|(D_n'D_n)^{-1}D_n'(A \otimes A)D_n|,$$

and evaluate the two determinants on the right-hand side separately.
First, consider $D_n'D_n$. The duplication matrix D_n has mutually orthogonal columns, each corresponding to a distinct element of an $n \times n$ symmetric matrix. For the n diagonal elements, the corresponding column in D_n

contains exactly one entry of 1, giving a squared norm of 1. For the $\frac{1}{2}n(n-1)$ off-diagonal elements, symmetry dictates that the corresponding column contains exactly two entries of 1, giving a squared norm of 2. Therefore, $D_n'D_n$ is a diagonal matrix with n ones and $\frac{1}{2}n(n-1)$ twos on its diagonal, yielding

$$|D_n'D_n| = 2^{\frac{1}{2}n(n-1)}.$$

Second, consider the linear operator $L = (D_n'D_n)^{-1}D_n'(A \otimes A)D_n$. For any symmetric matrix X, we have $D_n \operatorname{vech}(X) = \operatorname{vec} X$. Applying L gives

$$L \operatorname{vech}(X) = (D_n'D_n)^{-1}D_n'(A \otimes A) \operatorname{vec} X = (D_n'D_n)^{-1}D_n' \operatorname{vec}(AXA').$$

Since AXA' is symmetric, $(D_n'D_n)^{-1}D_n' \operatorname{vec}(AXA') = \operatorname{vech}(AXA')$. Thus, L acts on the half-vectorization space exactly as the transformation $X \mapsto AXA'$ acts on the space of symmetric matrices.
To find the determinant of L, we calculate its eigenvalues. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A with corresponding eigenvectors $v_1, \ldots, v_n$ (assuming A is diagonalizable). For $1 \leqslant j \leqslant i \leqslant n$, define the symmetric matrix $V_{ij} = v_i v_j' + v_j v_i'$. Applying the transformation yields

$$AV_{ij}A' = (Av_i)(Av_j)' + (Av_j)(Av_i)' = \lambda_i\lambda_j(v_i v_j' + v_j v_i') = \lambda_i\lambda_j V_{ij}.$$

The $\frac{1}{2}n(n+1)$ matrices V_{ij} are linearly independent and hence form a basis for the space of $n \times n$ symmetric matrices. The eigenvalues of L are therefore $\lambda_i\lambda_j$ $(1 \leqslant j \leqslant i \leqslant n)$. A specific eigenvalue λ_k appears $k-1$ times

when $i = k$ and $j < k$, $n - k$ times when $j = k$ and $i > k$, and contributes a power of 2 when $i = j = k$ (since $\lambda_k \lambda_k = \lambda_k^2$). Its total exponent is $(k-1) + (n-k) + 2 = n + 1$. Because this is true for all $k = 1, \ldots, n$,

$$|(D_n' D_n)^{-1} D_n'(A \otimes A) D_n| = \prod_{k=1}^{n} \lambda_k^{n+1} = |A|^{n+1}.$$

Multiplying the two determinants together completes the proof:

$$|D_n'(A \otimes A) D_n| = 2^{\frac{1}{2}n(n-1)} |A|^{n+1}.$$

(For non-diagonalizable A, the identity extends by continuity, since diagonalizable matrices are dense in $\mathbb{R}^{n \times n}$ and both sides are polynomial functions of the entries of A.) □

1.7 The determinant

Background needed for Jacobi's formula (32).

Definition 1. Let A be an $n \times n$ matrix. The *determinant* of A is defined as

$$|A| = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{1,\sigma(1)}\, a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}, \tag{7}$$

where S_n is the set of all $n!$ permutations of $\{1, 2, \ldots, n\}$, and $\operatorname{sgn}(\sigma) = +1$ if σ is an even permutation (expressible as a product of an even number of transpositions), and $\operatorname{sgn}(\sigma) = -1$ if σ is odd.

Consequence: $|A'| = |A|$. Indeed

$$|A'| = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{k} a_{\sigma(k),k} \overset{\rho=\sigma^{-1}}{=} \sum_{\rho} \operatorname{sgn}(\rho) \prod_{m} a_{m,\rho(m)} = |A|.$$

Proposition 11. *Interchanging two rows (or columns) of A multiplies the determinant by -1.*

Proof. Let B be obtained from A by swapping rows r and s ($r \neq s$), so that $b_{k,j} = a_{\tau(k),j}$ for all k, j, where $\tau = (r\ s)$ is the transposition of r and s. Then

$$|B| = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{k=1}^{n} a_{\tau(k),\sigma(k)}.$$

Substitute $k' = \tau(k)$. Since τ is a bijection and $\tau^{-1} = \tau$, as k ranges over $\{1, \ldots, n\}$ so does k', and $\sigma(k) =$

$\sigma(\tau(k'))$. Defining $\sigma' = \sigma \circ \tau$:

$$\prod_{k=1}^{n} a_{\tau(k),\sigma(k)} = \prod_{k'=1}^{n} a_{k',\sigma(\tau(k'))} = \prod_{k'=1}^{n} a_{k',\sigma'(k')}.$$

Since $\sigma \mapsto \sigma' = \sigma \circ \tau$ is a bijection on S_n and $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma' \circ \tau^{-1}) = \operatorname{sgn}(\sigma')\operatorname{sgn}(\tau) = -\operatorname{sgn}(\sigma')$:

$$|B| = \sum_{\sigma'} (-\operatorname{sgn}(\sigma')) \prod_{k'=1}^{n} a_{k',\sigma'(k')} = -|A|. \qquad \square$$

Corollary 1. *If two rows (or columns) of A are identical, then $|A| = 0$.*

Proof. Swapping the two identical rows leaves A unchanged, but multiplies $|A|$ by -1. Hence $|A| = -|A|$, giving $|A| = 0$. $\square$

Cofactor. (i, j)-minor M_{ij} = determinant after deleting row i, column j; cofactor $C_{ij} = (-1)^{i+j} M_{ij}$.

Proposition 12 (Laplace expansion)**.** *For any fixed row i:*

$$|A| = \sum_{j=1}^{n} a_{ij}\, C_{ij} = \sum_{j=1}^{n} (-1)^{i+j}\, a_{ij}\, M_{ij}. \tag{8}$$

Similarly, for any fixed column j:

$$|A| = \sum_{i=1}^{n} a_{ij}\, C_{ij}. \tag{9}$$

Proof. We prove the row expansion in two steps.
Step 1: Expansion along row 1. Group the terms in (7) by the value $j = \sigma(1)$:

$$|A| = \sum_{j=1}^{n} a_{1j} \sum_{\substack{\sigma \in S_n \\ \sigma(1)=j}} \operatorname{sgn}(\sigma) \prod_{k=2}^{n} a_{k,\sigma(k)}.$$

We must show that the inner sum equals $(-1)^{1+j} M_{1j}$.
Fix j and consider a permutation $\sigma \in S_n$ with $\sigma(1) = j$. The restriction of σ to $\{2, \ldots, n\}$ is a bijection onto $\{1, \ldots, n\} \setminus \{j\}$. Re-index this set as $\{1, \ldots, n-1\}$ via the order-preserving map ψ_j, and define $\tau \in S_{n-1}$ by

$\tau(r) = \psi_j(\sigma(r+1))$. The product becomes

$$\prod_{k=2}^{n} a_{k,\sigma(k)} = \prod_{r=1}^{n-1} \hat{a}_{r,\tau(r)},$$

where $\hat{a}_{r,s}$ are the entries of the $(n-1) \times (n-1)$ matrix $A^{(1j)}$ (row 1 and column j deleted). The map $\sigma \mapsto \tau$ is a bijection from $\{\sigma \in S_n : \sigma(1) = j\}$ to S_{n-1}.
The signs are related by $\operatorname{sgn}(\sigma) = (-1)^{j-1} \operatorname{sgn}(\tau)$. To see this, observe that σ can be reconstructed from τ in two stages: first, embed τ as a permutation $\hat{\tau} \in S_n$ that fixes position 1 and acts on positions $\{2, \ldots, n\}$ in the natural way (preserving sign); second, compose with the $j-1$ adjacent transpositions $(1\ 2)(2\ 3)\cdots(j{-}1\ j)$, which move the value j from position j to position 1. These $j-1$ transpositions contribute a sign of $(-1)^{j-1}$. Hence the inner sum equals

$$(-1)^{j-1} \sum_{\tau \in S_{n-1}} \operatorname{sgn}(\tau) \prod_{r=1}^{n-1} \hat{a}_{r,\tau(r)} = (-1)^{j-1} M_{1j}.$$

Since $(-1)^{j-1} = (-1)^{1+j}$, we obtain $|A| = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} M_{1j}$.
Step 2: General row i. Starting from A, swap row i with row $i-1$, then with row $i-2$, and so on, until it reaches row 1. This requires $i-1$ adjacent row swaps, so by Proposition 11:

$$|A| = (-1)^{i-1} |\tilde{A}|,$$

where $\tilde{A}$ has the original row i in position 1, with the remaining rows in their original relative order. The $(1, j)$-minor of $\tilde{A}$ equals M_{ij} (the (i, j)-minor of the original A), because the remaining rows below row 1 in $\tilde{A}$ are rows $1, \ldots, i-1, i+1, \ldots, n$ of A, in their original order. Applying Step 1 to $\tilde{A}$:

$$|\tilde{A}| = \sum_{j=1}^{n} (-1)^{1+j} \, a_{ij} \, M_{ij}.$$

Hence

$$|A| = (-1)^{i-1} \sum_{j=1}^{n} (-1)^{1+j} \, a_{ij} \, M_{ij} = \sum_{j=1}^{n} (-1)^{i+j} \, a_{ij} \, M_{ij}.$$

The column expansion (9) follows by applying (8) to A' and using $|A'| = |A|$. □

Two consequences:

Proposition 13. *For a nonsingular matrix A, the inverse is given by*

$$A^{-1} = \frac{1}{|A|} (C_{ji}),$$

where (C_{ji}) denotes the $n \times n$ matrix whose (i, j)-entry is C_{ji} (the cofactor matrix, transposed). Equivalently, $(A^{-1})_{ji} = C_{ij}/|A|$.

Proof. Define the matrix B with entries $B_{ij} = C_{ji}$. We need to show that $AB = |A|\, I$, i.e., $(AB)_{ik} = |A|\, \delta_{ik}$.

$$(AB)_{ik} = \sum_{j=1}^{n} a_{ij}\, B_{jk} = \sum_{j=1}^{n} a_{ij}\, C_{kj}.$$

If $i = k$: this is the Laplace expansion of $|A|$ along row i, so $(AB)_{ii} = |A|$. If $i \neq k$: this is the Laplace expansion along row k of a matrix $\tilde{A}$ that agrees with A except that row k has been replaced by row i. Since $\tilde{A}$ has two identical rows (i and k), $|\tilde{A}| = 0$ by Corollary 1. Hence $(AB)_{ik} = 0$. □

Corollary 2. *For any entry x_{ij} of a square matrix X:*

$$\frac{\partial |X|}{\partial x_{ij}} = C_{ij}.$$

Proof. From the Laplace expansion along row i: $|X| = \sum_k x_{ik} C_{ik}$. Since C_{ik} depends only on entries outside row i and column k, the cofactor C_{ij} does not depend on x_{ij}. Hence $\partial |X| / \partial x_{ij} = C_{ij}$. □

2 The definition of matrix derivative

Two pillars of matrix calculus: *derivative* (this section) and *differential* (next).

Linear case. If $f(x) = Ax$, the derivative is A:

$$\frac{\partial f(x)}{\partial x'} = A. \tag{10}$$

General $f : \mathbb{R}^n \to \mathbb{R}^m$.

$$\frac{\partial f(x)}{\partial x'} = \begin{pmatrix} \partial f_1/\partial x_1 & \dots & \partial f_1/\partial x_n \\ \vdots & & \vdots \\ \partial f_m/\partial x_1 & \dots & \partial f_m/\partial x_n \end{pmatrix} \qquad (m \times n). \tag{11}$$

- Row i: gradients of f_i. Column j: $\partial f/\partial x_j$.
- Scalar case: $\partial(a'x)/\partial x' = a'$, a *row* vector.

Matrix case. Order via vec:

$$DF(X) := \frac{\partial \operatorname{vec} F(X)}{\partial (\operatorname{vec} X)'}. \tag{12}$$

3 The differential

3.1 Definition, notation, and rules

For scalar f, the differential at x is the linear map

$$\mathrm{d}f(x)(u) = f'(x)u, \qquad u \in \mathbb{R}. \tag{13}$$

Differential (geometric) vs. derivative (algebraic representation).

Exercise 6. Show that

$$\begin{aligned} f(x) = x &\implies \mathrm{d}f(x)(u) = u, \\ f(x) = x^2 &\implies \mathrm{d}f(x)(u) = 2xu, \\ f(x) = \sin(x) &\implies \mathrm{d}f(x)(u) = \cos(x)\, u, \\ f(x) = \sin^2(x) &\implies \mathrm{d}f(x)(u) = 2\sin(x)\cos(x)\, u. \end{aligned}$$

Solution. These results follow directly from the definition in (13), where in the last case we apply the chain rule:

$$\frac{\mathrm{d}\sin^2(x)}{\mathrm{d}x} = \frac{\mathrm{d}\sin^2(x)}{\mathrm{d}\sin(x)} \cdot \frac{\mathrm{d}\sin(x)}{\mathrm{d}x} = 2\sin(x)\cos(x).$$

□

Identity $f(x) = x$ gives $\mathrm{d}x(u) = u$, so hereafter

$$\mathrm{d}f(x) = f'(x)\,\mathrm{d}x. \tag{14}$$

Exercise 7. Show that

$$\begin{aligned} \mathrm{d}x^2 &= 2x\,\mathrm{d}x, \\ \mathrm{d}\sin(x) &= \cos(x)\,\mathrm{d}x, \\ \mathrm{d}\sin^2(x) &= 2\sin(x)\cos(x)\,\mathrm{d}x. \end{aligned}$$

Solution. This follows immediately from the previous exercise by replacing u with $\mathrm{d}x$. □

Rules (linearity + Leibniz):

$$\begin{aligned} \mathrm{d}a &= 0, & \mathrm{d}(ax) &= a\,\mathrm{d}x & (a \text{ const.}), \\ \mathrm{d}(f+g) &= \mathrm{d}f + \mathrm{d}g, & \mathrm{d}(fg) &= (\mathrm{d}f)g + f\,\mathrm{d}g. \end{aligned} \tag{15}$$

3.2 Cauchy's rule of invariance

Chain rule, stated for differentials. If $z = f(g(x))$,

$$\mathrm{d}z = f'(y)\,\mathrm{d}y = f'(g(x))\,g'(x)\,\mathrm{d}x. \tag{16}$$

Exercise 8. Use Cauchy's rule of invariance to find the differentials of the functions $\sin^2(x)$, e^{x^2}, and $e^{\sin(x^2)}$.

Solution.

$$\begin{aligned} \mathrm{d}\sin^2(x) &= 2\sin(x)\,\mathrm{d}\sin(x) = 2\sin(x)\cos(x)\,\mathrm{d}x, \\ \mathrm{d}e^{x^2} &= e^{x^2}\,\mathrm{d}x^2 = 2e^{x^2}x\,\mathrm{d}x, \\ \mathrm{d}e^{\sin(x^2)} &= e^{\sin(x^2)}\cos(x^2)\,\mathrm{d}x^2 = 2x\,e^{\sin(x^2)}\cos(x^2)\,\mathrm{d}x. \end{aligned}$$

□

Identification. Reading the derivative off the differential:

$$\mathrm{d}f(x) = \alpha(x)\,\mathrm{d}x \iff f'(x) = \alpha(x). \tag{17}$$

(Special case of Prop. 14.)

3.3 Vector functions

Remark 2. Key: the differential preserves the function's order. $\mathrm{d}f$ is $m \times 1$ (like f), not $m \times n$ (like Df). This is why differentials scale from vectors to matrices.

For $f : \mathbb{R}^n \to \mathbb{R}^m$,

$$\mathrm{d}f(x)(u) = (Df(x))\,u, \tag{18}$$

and with $\mathrm{d}x(u) = u$,

$$\mathrm{d}f(x) = (Df(x))\,\mathrm{d}x. \tag{19}$$

Exercise 9. Find the differential of the function $f(x) = x_1^2 x_2$.

Solution. We have

$$\mathrm{d}f(x) = \frac{\partial f(x)}{\partial x_1}\,\mathrm{d}x_1 + \frac{\partial f(x)}{\partial x_2}\,\mathrm{d}x_2 = 2x_1x_2\,\mathrm{d}x_1 + x_1^2\,\mathrm{d}x_2,$$

where we note that both $f(x)$ and $\mathrm{d}f(x)$ are scalars. □

Rules generalize:

$$\begin{aligned} \mathrm{d}a &= 0, & \mathrm{d}(x') &= (\mathrm{d}x)', & \mathrm{d}(a'x) &= a'\,\mathrm{d}x, \\ \mathrm{d}(x+y) &= \mathrm{d}x + \mathrm{d}y, & \mathrm{d}(x'y) &= (\mathrm{d}x)'y + x'\,\mathrm{d}y. \end{aligned} \tag{20}$$

Goal: read off Df from $\mathrm{d}f$ (reverse of Exercise 9).

Exercise 10. Find again the differential of the function $f(x) = x_1^2 x_2$ without first calculating the partial derivatives.

Solution. Using the operating rules, we write

$$\mathrm{d}f(x) = \mathrm{d}(x_1^2 x_2) = (\mathrm{d}x_1^2)x_2 + x_1^2(\mathrm{d}x_2) = 2x_1x_2\,\mathrm{d}x_1 + x_1^2\,\mathrm{d}x_2 = (2x_1x_2,\ x_1^2)\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\end{pmatrix},$$

from which we find the derivative as $Df(x) = (2x_1x_2,\ x_1^2)$. □

3.4 First identification theorem

Generalizing (17):

Proposition 14 (First Identification Theorem)**.**

$$\mathrm{d}f(x) = A(x)\,\mathrm{d}x \iff Df(x) = A(x).$$

Proof. ($\Rightarrow$) By definition (19), $\mathrm{d}f(x) = (Df(x))\,\mathrm{d}x$. Since we also have $\mathrm{d}f(x) = A(x)\,\mathrm{d}x$, it follows that $(Df(x) - A(x))\,\mathrm{d}x = 0$ for all $\mathrm{d}x$. By Proposition 2(i), $Df(x) - A(x) = 0$.
($\Leftarrow$) If $Df(x) = A(x)$, then (19) gives $\mathrm{d}f(x) = A(x)\,\mathrm{d}x$ directly. □

Applications:

Exercise 11. Find the derivative of the linear function $a'x$, where a is a vector of constants, and of the quadratic function $x'Ax$, where A is a square matrix of constants.

Solution.

$$\begin{aligned}\mathrm{d}(a'x) &= a'\,\mathrm{d}x,\\ \mathrm{d}(x'Ax) &= (\mathrm{d}x)'Ax + x'A\,\mathrm{d}x = x'(A + A')\,\mathrm{d}x,\end{aligned}$$

using $(\mathrm{d}x)'Ax = x'A'\,\mathrm{d}x$ (scalar). Hence

$$\frac{\partial a'x}{\partial x'} = a', \qquad \frac{\partial x'Ax}{\partial x'} = x'(A + A').$$

(Row vectors; $2x'A$ when A is symmetric.) □

3.5 Cauchy invariance for vector functions

Chain rule:

$$\frac{\partial z}{\partial x'} = \frac{\partial z}{\partial y'}\frac{\partial y}{\partial x'}. \tag{21}$$

Proposition 15 (Cauchy Invariance)**.** *Let $z = f(y)$ and $y = g(x)$, so that $z = f(g(x))$. Then,*

$$\mathrm{d}z = A(y)B(x)\,\mathrm{d}x,$$

where $A(y)$ and $B(x)$ are defined through

$$\mathrm{d}z = A(y)\,\mathrm{d}y, \qquad \mathrm{d}y = B(x)\,\mathrm{d}x.$$

Proof. By (19), $\mathrm{d}z = (Dz(x))\,\mathrm{d}x$ and chain rule $Dz = Df(y)\,Dg(x)$. Prop. 14 gives $Df = A, Dg = B$, so $Dz = AB$ and $\mathrm{d}z = AB\,\mathrm{d}x$. □

Shorthand (conflating z and f):

$$\mathrm{d}f(x) = A(y)B(x)\,\mathrm{d}x, \tag{22}$$

with $\mathrm{d}f(y) = A(y)\,\mathrm{d}y$, $\mathrm{d}y = B(x)\,\mathrm{d}x$.

Exercise 12. Find the derivative of

$$f(x) = \begin{pmatrix} x_1^2 - x_2^2 \\ x_1x_2x_3 \end{pmatrix}, \qquad x = (x_1, x_2, x_3)'.$$

Solution. The differential is

$$\begin{aligned}\mathrm{d}f(x) &= \begin{pmatrix} \mathrm{d}(x_1^2) - \mathrm{d}(x_2^2) \\ \mathrm{d}(x_1x_2x_3) \end{pmatrix} = \begin{pmatrix} 2x_1\,\mathrm{d}x_1 - 2x_2\,\mathrm{d}x_2 \\ (\mathrm{d}x_1)x_2x_3 + x_1(\mathrm{d}x_2)x_3 + x_1x_2\,\mathrm{d}x_3 \end{pmatrix} \\ &= \begin{pmatrix} 2x_1 & -2x_2 & 0 \\ x_2x_3 & x_1x_3 & x_1x_2 \end{pmatrix} \begin{pmatrix} \mathrm{d}x_1 \\ \mathrm{d}x_2 \\ \mathrm{d}x_3 \end{pmatrix},\end{aligned}$$

which identifies the derivative as

$$\frac{\partial f(x)}{\partial x'} = \begin{pmatrix} 2x_1 & -2x_2 & 0 \\ x_2x_3 & x_1x_3 & x_1x_2 \end{pmatrix}.$$

□

Exercise 13. Let

$$f(y) = e^{y_1}\sin(y_2), \qquad y_1 = x_1 x_2^2, \qquad y_2 = x_1^2 x_2.$$

Find the derivative of f with respect to $x = (x_1, x_2)'$.

Solution. Using the notational convention (22), we write

$$\mathrm{d}f(y) = (\mathrm{d}e^{y_1})\sin(y_2) + e^{y_1}\,\mathrm{d}\sin(y_2) = e^{y_1}\sin(y_2)\,\mathrm{d}y_1 + e^{y_1}\cos(y_2)\,\mathrm{d}y_2 = a(y)'\,\mathrm{d}y,$$

where

$$a(y) = e^{y_1}\begin{pmatrix}\sin(y_2)\\ \cos(y_2)\end{pmatrix}, \qquad \mathrm{d}y = \begin{pmatrix}\mathrm{d}y_1\\ \mathrm{d}y_2\end{pmatrix}.$$

Also,

$$\mathrm{d}y = \begin{pmatrix} x_2^2 & 2x_1x_2 \\ 2x_1x_2 & x_1^2\end{pmatrix}\begin{pmatrix}\mathrm{d}x_1\\ \mathrm{d}x_2\end{pmatrix} = B(x)\,\mathrm{d}x.$$

Hence,

$$\mathrm{d}f(x) = a(y)'\,\mathrm{d}y = a(y)'B(x)\,\mathrm{d}x = c_1\,\mathrm{d}x_1 + c_2\,\mathrm{d}x_2,$$

where

$$\begin{aligned} c_1 &= x_2 e^{y_1}\big(x_2\sin(y_2) + 2x_1\cos(y_2)\big),\\ c_2 &= x_1 e^{y_1}\big(x_1\cos(y_2) + 2x_2\sin(y_2)\big), \end{aligned}$$

so that the derivative is $\partial f(x)/\partial x' = (c_1, c_2)$. □

4 Optimization

Unconstrained. Compute $\mathrm{d}f(x) = a(x)'\,\mathrm{d}x$, solve $a(x) = 0$.

Exercise 14. Minimize the function

$$f(x) = \tfrac{1}{2}x'Ax - b'x,$$

where the matrix A is positive definite.

Solution. The differential is

$$\mathrm{d}f(x) = x'A\,\mathrm{d}x - b'\,\mathrm{d}x = (Ax - b)'\,\mathrm{d}x,$$

since a positive definite matrix is symmetric, by definition. The solution $\hat{x}$ needs to satisfy $A\hat{x} - b = 0$, and hence $\hat{x} = A^{-1}b$. The function f has an absolute minimum at $\hat{x}$, which can be seen by defining $y = x - \hat{x}$ and writing

$$y'Ay = (x - A^{-1}b)'A(x - A^{-1}b) = 2f(x) + b'A^{-1}b.$$

Since A is positive definite, $y'Ay$ has a minimum at $y = 0$ and hence $f(x)$ has a minimum at $x = \hat{x}$. Alternatively, we note that $f(x)$ is strictly convex and use the fact that any (strictly) convex function attains a (strict) absolute minimum. □

Scalar constraint $g(x) = 0$**.** Lagrangian

$$\mathcal{L}(x) = f(x) - \lambda g(x), \tag{23}$$

$$\mathrm{d}\mathcal{L}(x) = \mathrm{d}f(x) - \lambda\,\mathrm{d}g(x) = 0. \tag{24}$$

First-order conditions ($n+1$ unknowns x, λ):

$$\frac{\partial f}{\partial x'} = \lambda \frac{\partial g}{\partial x'}, \qquad g(x) = 0. \tag{25}$$

Exercise 15. Let A be positive definite. Minimize the function

$$f(x) = \tfrac{1}{2} x' A x - b' x,$$

subject to the constraint $c'x = 0$.

Solution. We write the Lagrangian as

$$\mathcal{L}(x) = \tfrac{1}{2} x' A x - b' x - \lambda c' x.$$

The differential of $\mathcal{L}(x)$ is

$$\mathrm{d}\mathcal{L}(x) = (Ax - b)' \,\mathrm{d}x - \lambda c' \,\mathrm{d}x = (Ax - b - \lambda c)' \,\mathrm{d}x,$$

from which we obtain the first-order conditions $Ax - b - \lambda c = 0$ and $c'x = 0$. This gives $Ax = b + \lambda c$, and isolating x yields $x = A^{-1}(b + \lambda c)$. We then substitute this expression for x into the constraint equation $c'x = 0$:

$$0 = c' A^{-1}(b + \lambda c) = c' A^{-1} b + \lambda c' A^{-1} c.$$

Notice that $c'A^{-1}b$ and $c'A^{-1}c$ are both scalars. Since A is positive definite, $c'A^{-1}c > 0$, allowing us to isolate the Lagrange multiplier λ:

$$\lambda c'A^{-1}c = -c'A^{-1}b \implies \tilde{\lambda} = -\frac{c'A^{-1}b}{c'A^{-1}c}.$$

Substituting $\tilde{\lambda}$ back into the equation for x yields the constrained optimum:

$$\tilde{x} = A^{-1}(b + \tilde{\lambda}c) = A^{-1}b - \frac{c'A^{-1}b}{c'A^{-1}c}A^{-1}c.$$

Alternatively, we can write the first-order conditions as

$$\begin{pmatrix} A & c \\ c' & 0 \end{pmatrix} \begin{pmatrix} x \\ -\lambda \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}$$

and solve for x and $-\lambda$ by inverting the matrix on the left-hand side. Notice that by expressing the equation in x and $-\lambda$ (rather than x and λ) we achieve symmetry of the matrix on the left-hand side. Symmetric matrices are easier, both theoretically and computationally, than non-symmetric matrices, so, if possible, we work with symmetric matrices. Either way, since $f(x)$ is linear-quadratic (hence strictly convex) and the constraint is linear, $f(x)$ attains an absolute minimum at $\tilde{x}$ under the constraint. □

Vector constraint $g(x) = 0$ (m equations). With $l = (\lambda_1, \ldots, \lambda_m)'$:

$$\mathcal{L}(x) = f(x) - l'g(x), \tag{26}$$

$$\mathrm{d}\mathcal{L}(x) = \mathrm{d}f(x) - l'\,\mathrm{d}g(x) = 0. \tag{27}$$

$$\frac{\partial f}{\partial x'} = l'\frac{\partial g}{\partial x'}, \qquad g(x) = 0. \tag{28}$$

Exercise 16. Show that optimizing a function subject to constraints is not equivalent to optimizing the Lagrangian function.

Solution. A simple counterexample is provided by the optimization problem: $\max(xy)$ under the constraint $x + y = 2$. The solution is $x = y = 1$, but the Lagrangian $\mathcal{L}(x, y) = xy - \lambda(x + y - 2)$ has a saddle-point at $(1, 1)$. □

Remark 3. FOCs are necessary. When $\mathcal{L}$ is (strictly) convex (Exercises 14, 15), $\hat{x}$ is a (strict) constrained minimum. In Exercise 16, $H\mathcal{L} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is indefinite, so convexity fails.

5 Example 1: Least squares

Given A ($n \times k$, $\mathrm{r}\,A = k$) and b. Let $e = b - Ax$. Minimize

$$f(x) = \tfrac{1}{2}\,e'e = \tfrac{1}{2}(b - Ax)'(b - Ax). \tag{29}$$

(The 1/2 cancels the 2 from differentiation.) Differentiating,

$$\mathrm{d}f(x) = e'\,\mathrm{d}e = -e'A\,\mathrm{d}x.$$

So $A'e = 0$, i.e. $A'Ax = A'b$:

$$\hat{x} = (A'A)^{-1}A'b. \tag{30}$$

Constrained LS: $Rx = r$ (R full row rank). Lagrangian

$$\mathcal{L}(x) = \tfrac{1}{2}(b - Ax)'(b - Ax) - l'(Rx - r).$$

Exercise 17. Show that the matrix $V = R(A'A)^{-1}R'$ is nonsingular if and only if the m rows of R are linearly independent.

Solution. Let $B = R(A'A)^{-1/2}$, so that $\mathrm{r}(B) = \mathrm{r}(R)$. Since $V = BB'$, it follows that $\mathrm{r}(V) = \mathrm{r}(B)$, and hence that $\mathrm{r}(V) = \mathrm{r}(R)$. Since V is an $m \times m$ matrix, it is nonsingular if and only if $\mathrm{r}(V) = m$, that is, if and only if $\mathrm{r}(R) = m$. □

The differential

$$\mathrm{d}\mathcal{L}(x) = -(b - Ax)'A\,\mathrm{d}x - l'R\,\mathrm{d}x$$

yields FOCs $A'A\tilde{x} - A'b = R'\tilde{l}$, $R\tilde{x} = r$. Premultiplying by $R(A'A)^{-1}$ and using $\hat{x}$ from (30):

$$\tilde{l} = \big(R(A'A)^{-1}R'\big)^{-1}(r - R\hat{x}),$$

$$\tilde{x} = \hat{x} + (A'A)^{-1}R'\big(R(A'A)^{-1}R'\big)^{-1}(r - R\hat{x}). \tag{31}$$

Strictly convex quadratic + linear constraint $\Rightarrow \tilde{x}$ is the constrained minimum.

6 Matrix calculus

Matrix differential rules (A constant, α scalar):

$$\mathrm{d}A = 0, \quad \mathrm{d}(\alpha X) = \alpha\,\mathrm{d}X, \quad \mathrm{d}(X') = (\mathrm{d}X)', \quad \mathrm{d}\,\mathrm{tr}\,X = \mathrm{tr}\,\mathrm{d}X,$$

$$\mathrm{d}(X+Y) = \mathrm{d}X + \mathrm{d}Y, \qquad \mathrm{d}(XY) = (\mathrm{d}X)Y + X\,\mathrm{d}Y.$$

Non-trivial (nonsingular X):

$$\mathrm{d}|X| = |X|\ \mathrm{tr}\,X^{-1}\,\mathrm{d}X, \tag{32}$$

$$\mathrm{d}\log|X| = \mathrm{tr}\,X^{-1}\,\mathrm{d}X \qquad (|X| > 0). \tag{33}$$

Proof of (32). By Corollary 2, $\partial|X|/\partial x_{ij} = C_{ij}$. Hence

$$\mathrm{d}|X| = \sum_{i,j} C_{ij}\,\mathrm{d}x_{ij}.$$

By Proposition 13, $C_{ij} = |X|\,(X^{-1})_{ji}$. Substituting:

$$\begin{aligned}\mathrm{d}|X| &= \sum_{i,j} |X|\,(X^{-1})_{ji}\,\mathrm{d}x_{ij} = |X| \sum_j \sum_i (X^{-1})_{ji}\,\mathrm{d}x_{ij} \\ &= |X| \sum_j (X^{-1}\,\mathrm{d}X)_{jj} = |X|\ \mathrm{tr}\,X^{-1}\,\mathrm{d}X.\end{aligned}$$

□

Inverse. For nonsingular X,

$$\mathrm{d}X^{-1} = -X^{-1}(\mathrm{d}X)X^{-1}. \tag{34}$$

(Differentiate $X^{-1}X = I$ and postmultiply by X^{-1}.)

Matrix identification.

$$\mathrm{d}\,\mathrm{vec}\,F(X) = A(X)\,\mathrm{d}\,\mathrm{vec}\,X \iff DF(X) = A(X). \tag{35}$$

Chain rule: $Z = F(G(X))$ gives $\mathrm{d}Z = A(Y)B(X)\,\mathrm{d}X$ (Prop. 15).

Matrix constraints $G(X) = 0$. With multipliers $L = (\lambda_{ij})$:

$$\mathcal{L}(X) = f(X) - \mathrm{tr}\,L'G(X) = f(X) - \sum_{ij} \lambda_{ij} g_{ij}. \tag{36}$$

If G symmetric, take L symmetric.

Exercise 18. Consider the optimization problem

$$\text{minimize } f(X) \quad \text{subject to } G(X) = 0.$$

If $G(X)$ is symmetric for all X, then the Lagrangian function is

$$\mathcal{L}(X) = f(X) - \mathrm{tr}\,LG(X),$$

where L may be assumed to be symmetric.

Solution. Since G is symmetric, we have

$$\operatorname{tr} L'G = \operatorname{tr} L'G' = \operatorname{tr}(GL)' = \operatorname{tr} GL = \operatorname{tr} LG,$$

so that $\operatorname{tr} L'G = \operatorname{tr} L^*G$, where $L^* = (L + L')/2$ is symmetric. □

From (36):

$$\mathrm{d}\mathcal{L}(X) = \mathrm{d}f(X) - \operatorname{tr} L' \,\mathrm{d}G(X) = 0. \tag{37}$$

FOCs (using Prop. 5):

$$\frac{\partial f}{\partial (\operatorname{vec} X)'} = (\operatorname{vec} L)' \frac{\partial \operatorname{vec} G}{\partial (\operatorname{vec} X)'}, \qquad G(X) = 0. \tag{38}$$

7 Four exercises: first derivative

Exercise 19. Obtain the derivative of the scalar function $f(X) = \operatorname{tr} X'AX$.

Solution. The differential is

$$\begin{aligned} \mathrm{d}f(X) &= \mathrm{d}(\operatorname{tr} X'AX) = \operatorname{tr} \mathrm{d}(X'AX) \\ &= \operatorname{tr}(\mathrm{d}X)'AX + \operatorname{tr} X'A \,\mathrm{d}X. \end{aligned}$$

To combine these terms, we use the property that the trace of a matrix equals the trace of its transpose

($\operatorname{tr} M = \operatorname{tr} M'$). Applying this to the first term gives:

$$\operatorname{tr}\big((\mathrm{d}X)'AX\big) = \operatorname{tr}\big(((\mathrm{d}X)'AX)'\big) = \operatorname{tr}(X'A'\,\mathrm{d}X).$$

Substituting this back into the differential, we can factor out X' and $\mathrm{d}X$:

$$\begin{aligned}\mathrm{d}f(X) &= \operatorname{tr} X'A'\,\mathrm{d}X + \operatorname{tr} X'A\,\mathrm{d}X \\ &= \operatorname{tr} X'(A + A')\,\mathrm{d}X \\ &= \operatorname{tr} C'\,\mathrm{d}X = (\operatorname{vec} C)'\,\mathrm{d}\operatorname{vec} X,\end{aligned}$$

using Proposition 5 and letting $C = (A + A')X$. Hence the derivative is $Df(X) = (\operatorname{vec} C)'$. □

Exercise 20. Obtain the derivative of the scalar function $f(X) = \log|X'X|$, where X has full column rank.

Solution. From the differential

$$\begin{aligned}\mathrm{d}f(X) &= \mathrm{d}\log|X'X| = \operatorname{tr}(X'X)^{-1}\mathrm{d}(X'X) \\ &= \operatorname{tr}(X'X)^{-1}(\mathrm{d}X)'X + \operatorname{tr}(X'X)^{-1}X'\,\mathrm{d}X = 2\operatorname{tr}(X'X)^{-1}X'\,\mathrm{d}X \\ &= 2\operatorname{tr} C'\,\mathrm{d}X = 2(\operatorname{vec} C)'\,\mathrm{d}\operatorname{vec} X,\end{aligned}$$

where $C = X(X'X)^{-1}$, we obtain $Df(X) = 2(\operatorname{vec} C)'$. □

Exercise 21. Let $f_k(X) = \operatorname{tr} X^k$ $(k = 1, 2, \dots)$. Find the derivative.

Solution. We have

$$\begin{aligned} \mathrm{d}f_k(X) &= \operatorname{tr}(\mathrm{d}X)X^{k-1} + \operatorname{tr} X(\mathrm{d}X)X^{k-2} + \cdots + \operatorname{tr} X^{k-1}\,\mathrm{d}X \\ &= k \operatorname{tr} X^{k-1}\,\mathrm{d}X = k(\operatorname{vec} X'^{k-1})'\,\mathrm{d}\operatorname{vec} X. \end{aligned}$$

This gives $Df_k(X) = k(\operatorname{vec} X'^{k-1})'$. In particular,

$$Df_1(X) = D\operatorname{tr} X = (\operatorname{vec} I)', \qquad Df_2(X) = D\operatorname{tr} X^2 = 2(\operatorname{vec} X')'.$$

□

Exercise 22. Find the derivative of the matrix equation $F(X) = AX^{-1}B$, where X is nonsingular.

Solution. Since

$$\mathrm{d}F(X) = A(\mathrm{d}X^{-1})B = -AX^{-1}(\mathrm{d}X)X^{-1}B,$$

we need to vectorize this equation to find the derivative. We use Proposition 6, which states that $\operatorname{vec}(M_1M_2M_3) = (M_3' \otimes M_1)\operatorname{vec} M_2$ for any compatible matrices.
By grouping the terms in our differential as $M_1 = -AX^{-1}$, $M_2 = \mathrm{d}X$, and $M_3 = X^{-1}B$, we find:

$$\mathrm{d}\operatorname{vec} F(X) = \operatorname{vec}\big((-AX^{-1})(\mathrm{d}X)(X^{-1}B)\big) = -\big((X^{-1}B)' \otimes (AX^{-1})\big)\,\mathrm{d}\operatorname{vec} X.$$

The derivative is thus identified by the first identification theorem for matrices (35) as

$$DF(X) = \frac{\partial \operatorname{vec} F(X)}{\partial (\operatorname{vec} X)'} = -(X^{-1}B)' \otimes (AX^{-1}).$$

□

8 The second differential

$\mathrm{d}^2 f := \mathrm{d}(\mathrm{d}f)$.

Exercise 23. Let $f(x) = x'Ax$. Show that $\mathrm{d}^2 f(x) = (\mathrm{d}x)'(A + A')\,\mathrm{d}x$.

Solution. We know from Exercise 11 that $\mathrm{d}f(x) = x'(A + A')\,\mathrm{d}x$. Then,

$$\mathrm{d}^2 f(x) = d\big(x'(A + A')\,\mathrm{d}x\big) = (\mathrm{d}x)'(A + A')\,\mathrm{d}x + x'(A + A')\,\mathrm{d}^2 x = (\mathrm{d}x)'(A + A')\,\mathrm{d}x,$$

since $\mathrm{d}^2 x = 0$. □

Exercise 24. Why is $\mathrm{d}^2 x = 0$?

Solution. $\mathrm{d}x$ is the identity differential $\mathrm{d}x(u) = u$ (Section 3.1), whose second derivative is 0. Hence $\mathrm{d}^2 x = 0$ when x is the independent variable. If $x = x(t)$, then $\mathrm{d}^2 x \neq 0$ (unless x is linear in t); see Section 9. □

Hessian is defined only for scalar f (vector/matrix case: componentwise).

Let f be scalar, and

$$\mathrm{d}f(x) = a(x)'\,\mathrm{d}x, \qquad \mathrm{d}a(x) = (Hf(x))\,\mathrm{d}x, \tag{39}$$

where

$$a(x)' = \frac{\partial f(x)}{\partial x'}, \qquad Hf(x) = \frac{\partial a(x)}{\partial x'} = \frac{\partial}{\partial x'}\left(\frac{\partial f(x)}{\partial x'}\right)'.$$

$(Hf)_{ij} = \partial^2 f/\partial x_i \partial x_j$, symmetric (twice differentiable f). Standard notation:

$$Hf(x) = \frac{\partial^2 f(x)}{\partial x\,\partial x'}, \tag{40}$$

$$\frac{\partial^2 f(x)}{\partial x\,\partial x'} = \frac{\partial}{\partial x'}\left(\frac{\partial f(x)}{\partial x'}\right)'. \tag{41}$$

Second differential is a quadratic form:

$$\mathrm{d}^2 f(x) = (\mathrm{d}x)'\,Hf(x)\,\mathrm{d}x. \tag{42}$$

If $\mathrm{d}^2 f(x) = (\mathrm{d}x)'B(x)\,\mathrm{d}x$ with B not necessarily symmetric, symmetry of Hf gives $Hf = (B + B')/2$ (Prop. 2(ii)):

Proposition 16 (Second Identification Theorem)**.**

$$\mathrm{d}^2 f(x) = (\mathrm{d}x)'B(x)\,\mathrm{d}x \iff Hf(x) = \frac{B(x) + B(x)'}{2}.$$

Exercise 25. Consider again the quadratic function $f(x) = x'Ax$. Find the second differential without writing out the first differential in its final form.

Solution. From Exercise 23, $\mathrm{d}^2 f = (\mathrm{d}x)'(A + A')\,\mathrm{d}x$, already symmetric, so $Hf = A + A'$. Shortcut: from $\mathrm{d}f = (\mathrm{d}x)'Ax + x'A\,\mathrm{d}x$ directly take $\mathrm{d}^2 f = 2(\mathrm{d}x)'A\,\mathrm{d}x$, giving $Hf = A + A'$. □

9 Chain rule for second differentials

If $z = f(y)$, $y = g(x)$, then $\mathrm{d}z = A(y)\,\mathrm{d}y$ and

$$\mathrm{d}^2 z = (\mathrm{d}A)\,\mathrm{d}y + A(y)\,\mathrm{d}^2 y, \tag{43}$$

with $\mathrm{d}y = B(x)\,\mathrm{d}x$ and $\mathrm{d}^2 y = (\mathrm{d}B)\,\mathrm{d}x$ (not zero when y is intermediate).

Proposition 17 (Chain Rule for Second Differentials)**.** *Let $z = f(y)$ and $y = g(x)$, so that $z = f(g(x))$. Then,*

$$\mathrm{d}^2 z = (\mathrm{d}A)B(x)\,\mathrm{d}x + A(y)(\mathrm{d}B)\,\mathrm{d}x,$$

where $A(y)$ and $B(x)$ are defined through

$$\mathrm{d}z = A(y)\,\mathrm{d}y, \qquad \mathrm{d}y = B(x)\,\mathrm{d}x.$$

Proof. Product rule on $\mathrm{d}z = A(y)\,\mathrm{d}y$: $\mathrm{d}^2 z = (\mathrm{d}A)\,\mathrm{d}y + A\,\mathrm{d}^2 y$. Using $\mathrm{d}y = B\,\mathrm{d}x$, $\mathrm{d}^2 x = 0$, we get $\mathrm{d}^2 y = (\mathrm{d}B)\,\mathrm{d}x$, whence the result. □

Remark 4. In practice one often uses (43) directly instead of Prop. 17. Two examples below.

Exercise 26. Let

$$f(y_1, y_2) = e^{y_1} \sin(y_2), \qquad y_1 = x_1 x_2^2, \qquad y_2 = x_1^2 x_2.$$

Find the Hessian matrix by using Proposition 17.

Solution. By Proposition 17,

$$\mathrm{d}^2 f(x) = (\mathrm{d}a)' B(x)\,\mathrm{d}x + a(y)'(\mathrm{d}B)\,\mathrm{d}x,$$

where

$$a(y) = e^{y_1} \begin{pmatrix} \sin(y_2) \\ \cos(y_2) \end{pmatrix}, \qquad B(x) = \begin{pmatrix} x_2^2 & 2x_1x_2 \\ 2x_1x_2 & x_1^2 \end{pmatrix}.$$

Now, letting

$$C(y) = e^{y_1} \begin{pmatrix} \sin(y_2) & \cos(y_2) \\ \cos(y_2) & -\sin(y_2) \end{pmatrix}$$

and

$$D_1(x) = 2 \begin{pmatrix} 0 & x_2 \\ x_2 & x_1 \end{pmatrix}, \qquad D_2(x) = 2 \begin{pmatrix} x_2 & x_1 \\ x_1 & 0 \end{pmatrix},$$

we obtain

$$\mathrm{d}a = C(y)\,\mathrm{d}y = C(y)B(x)\,\mathrm{d}x$$

and

$$\mathrm{d}B = (\mathrm{d}x_1)D_1(x) + (\mathrm{d}x_2)D_2(x).$$

Let us write $\mathrm{d}x_1$ and $\mathrm{d}x_2$ in terms of $\mathrm{d}x$, which can be done by defining $e_1 = (1,0)'$ and $e_2 = (0,1)'$. Then, $\mathrm{d}x_1 = e_1'\,\mathrm{d}x$ and $\mathrm{d}x_2 = e_2'\,\mathrm{d}x$, and hence

$$\begin{aligned}
\mathrm{d}^2 f(x) &= (\mathrm{d}a)'B(x)\,\mathrm{d}x + a(y)'(\mathrm{d}B)\,\mathrm{d}x \\
&= (\mathrm{d}x)'B(x)C(y)B(x)\,\mathrm{d}x + a(y)'\big((\mathrm{d}x_1)D_1(x) + (\mathrm{d}x_2)D_2(x)\big)\,\mathrm{d}x \\
&= (\mathrm{d}x)'B(x)C(y)B(x)\,\mathrm{d}x + (\mathrm{d}x)'e_1 a(y)'D_1(x)\,\mathrm{d}x + (\mathrm{d}x)'e_2 a(y)'D_2(x)\,\mathrm{d}x \\
&= (\mathrm{d}x)'\big[B(x)C(y)B(x) + e_1 a(y)'D_1(x) + e_2 a(y)'D_2(x)\big]\,\mathrm{d}x.
\end{aligned}$$

Some care is required where to position the scalars $e_1'\,\mathrm{d}x$ and $e_2'\,\mathrm{d}x$ in the matrix product. A scalar can be positioned anywhere in a matrix product, but we wish to position the two scalars in such a way that the usual matrix multiplication rules still apply. Having obtained the second differential in the desired form, Proposition 16 implies that the Hessian is equal to

$$\begin{aligned}
Hf(x) = B(x)C(y)B(x) &\\
&+ \tfrac{1}{2}\big(e_1 a(y)'D_1(x) + D_1(x)a(y)e_1'\big) \\
&+ \tfrac{1}{2}\big(e_2 a(y)'D_2(x) + D_2(x)a(y)e_2'\big).
\end{aligned}$$

□

Exercise 27. Now find the Hessian matrix, not using Proposition 17.

Solution. We have

$$\mathrm{d}f(y) = (\mathrm{d}e^{y_1})\sin(y_2) + e^{y_1}\,\mathrm{d}\sin(y_2) = e^{y_1}\sin(y_2)\,\mathrm{d}y_1 + e^{y_1}\cos(y_2)\,\mathrm{d}y_2,$$

and hence

$$\begin{aligned}\mathrm{d}^2 f(y) &= [\mathrm{d}e^{y_1}]\sin(y_2)\,\mathrm{d}y_1 + e^{y_1}[\mathrm{d}\sin(y_2)]\,\mathrm{d}y_1 + e^{y_1}\sin(y_2)\,\mathrm{d}^2 y_1 \\ &\quad + [\mathrm{d}e^{y_1}]\cos(y_2)\,\mathrm{d}y_2 + e^{y_1}[\mathrm{d}\cos(y_2)]\,\mathrm{d}y_2 + e^{y_1}\cos(y_2)\,\mathrm{d}^2 y_2 \\ &= e^{y_1}\sin(y_2)(\mathrm{d}y_1)^2 + e^{y_1}\cos(y_2)\,\mathrm{d}y_1\,\mathrm{d}y_2 + e^{y_1}\sin(y_2)\,\mathrm{d}^2 y_1 \\ &\quad + e^{y_1}\cos(y_2)\,\mathrm{d}y_1\,\mathrm{d}y_2 - e^{y_1}\sin(y_2)(\mathrm{d}y_2)^2 + e^{y_1}\cos(y_2)\,\mathrm{d}^2 y_2,\end{aligned}$$

where we emphasize that $\mathrm{d}^2 y_1$ and $\mathrm{d}^2 y_2$ are not 0, because they depend on x_1 and x_2. In fact,

$$\mathrm{d}y_1 = x_2^2\,\mathrm{d}x_1 + 2x_1x_2\,\mathrm{d}x_2, \qquad \mathrm{d}^2 y_1 = 4x_2\,\mathrm{d}x_1\,\mathrm{d}x_2 + 2x_1(\mathrm{d}x_2)^2,$$

and

$$\mathrm{d}y_2 = 2x_1x_2\,\mathrm{d}x_1 + x_1^2\,\mathrm{d}x_2, \qquad \mathrm{d}^2 y_2 = 2x_2(\mathrm{d}x_1)^2 + 4x_1\,\mathrm{d}x_1\,\mathrm{d}x_2.$$

Inserting these expressions into $\mathrm{d}^2 f(y)$ gives the required result. □

10 The Hessian matrix

Viewing $F(X)$ as $f = \operatorname{vec} F$, $x = \operatorname{vec} X$ (cf. (35)), the second identification theorem becomes

$$\mathrm{d}^2 f(X) = (\mathrm{d}\operatorname{vec} X)' B(X)\,\mathrm{d}\operatorname{vec} X \iff Hf(X) = \frac{B(X) + B(X)'}{2}. \tag{44}$$

(Scalar f only.) Second differentials often appear as traces $\operatorname{tr} A(\mathrm{d}X)'B\,\mathrm{d}X$ or $\operatorname{tr} A(\mathrm{d}X)B\,\mathrm{d}X$:

Proposition 18. *Let f be a twice differentiable real-valued function of an $n \times q$ matrix X. Then,*

$$\mathrm{d}^2 f(X) = \operatorname{tr} A(\mathrm{d}X)'B\,\mathrm{d}X \iff Hf(X) = \tfrac{1}{2}(A' \otimes B + A \otimes B'),$$
$$\mathrm{d}^2 f(X) = \operatorname{tr} A(\mathrm{d}X)B\,\mathrm{d}X \iff Hf(X) = \tfrac{1}{2}K_{qn}(A' \otimes B + B' \otimes A).$$

Proof. We use the trace–vec identity $\operatorname{tr} M'N = (\operatorname{vec} M)' \operatorname{vec} N$ (Proposition 5) and the identity $\operatorname{vec}(PQR) = (R' \otimes P)\operatorname{vec} Q$ (Proposition 6), along with $K_{nq}\operatorname{vec} X = \operatorname{vec} X'$ (definition of the commutation matrix).
First identity. Cyclically permuting the trace and applying the identities:

$$\begin{aligned}\operatorname{tr} A(\mathrm{d}X)'B\,\mathrm{d}X &= \operatorname{tr}(\mathrm{d}X)'B(\mathrm{d}X)A = (\operatorname{vec}\mathrm{d}X)'\operatorname{vec} B(\mathrm{d}X)A \\ &= (\operatorname{vec}\mathrm{d}X)'(A' \otimes B)\operatorname{vec}\mathrm{d}X = (\mathrm{d}\operatorname{vec} X)'(A' \otimes B)\,\mathrm{d}\operatorname{vec} X.\end{aligned}$$

By the second identification theorem (Proposition 16), $Hf(X) = \frac{1}{2}((A' \otimes B) + (A' \otimes B)') = \frac{1}{2}(A' \otimes B + A \otimes B')$.

Second identity. Using $\operatorname{tr} M = \operatorname{tr} M'$, we rewrite $\operatorname{tr} A(\mathrm{d}X)B\,\mathrm{d}X = \operatorname{tr}(\mathrm{d}X)'B'(\mathrm{d}X)'A'$:

$$\begin{aligned}\operatorname{tr} A(\mathrm{d}X)B\,\mathrm{d}X &= \operatorname{tr}(\mathrm{d}X)'B'(\mathrm{d}X)'A' = (\operatorname{vec}\mathrm{d}X)'\operatorname{vec} B'(\mathrm{d}X)'A' \\ &= (\operatorname{vec}\mathrm{d}X)'(A\otimes B')\operatorname{vec}(\mathrm{d}X)' = (\mathrm{d}\operatorname{vec} X)'(A\otimes B')K_{nq}\,\mathrm{d}\operatorname{vec} X.\end{aligned}$$

Applying Proposition 16, $Hf(X) = \frac{1}{2}\big((A\otimes B')K_{nq} + K_{qn}(A\otimes B')'\big) = \frac{1}{2}\big((A\otimes B')K_{nq} + K_{qn}(A'\otimes B)\big)$, using $K_{nq}' = K_{qn}$ and $(A\otimes B')' = A'\otimes B$.
It remains to simplify this last expression. Applying Proposition 8 with the roles of A and B swapped (i.e., $K_{pm}(A\otimes B) = (B\otimes A)K_{qn}$, read right-to-left) gives $(A\otimes B')K_{nq} = K_{qn}(B'\otimes A)$. Therefore

$$(A\otimes B')K_{nq} + K_{qn}(A'\otimes B) = K_{qn}(B'\otimes A) + K_{qn}(A'\otimes B) = K_{qn}(A'\otimes B + B'\otimes A),$$

which yields $Hf(X) = \frac{1}{2}K_{qn}(A'\otimes B + B'\otimes A)$. □

Exercise 28. Suppose that $\mathrm{d}^2 f(X) = 2\operatorname{tr} A(\mathrm{d}X)'A\,\mathrm{d}X$ and that both A and X are known to be symmetric $n\times n$ matrices. Show that

$$Hf(X) = 2D_n'(A\otimes A)D_n.$$

Solution. This follows from

$$\operatorname{tr} A(\mathrm{d}X)'A\,\mathrm{d}X = (\mathrm{d}\operatorname{vec} X)'(A\otimes A)\,\mathrm{d}\operatorname{vec} X = (\mathrm{d}\operatorname{vech}(X))'D_n'(A\otimes A)D_n\,\mathrm{d}\operatorname{vech}(X),$$

using symmetry of A so that $A' = A$ in Proposition 18, and $\mathrm{d}\,\mathrm{vec}\,X = D_n\,\mathrm{d}\,\mathrm{vech}(X)$ by (5). □

11 Four exercises: second derivative

Revisiting Section 7 for Hessians.

Exercise 29 (Exercise 19 Cont'd)**.** Obtain the Hessian of $f(X) = \mathrm{tr}\,X'AX$.

Solution. We know that $\mathrm{d}f(X) = \mathrm{tr}\,C'\,\mathrm{d}X$, where $C = (A + A')X$. The second differential is

$$\mathrm{d}^2 f(X) = \mathrm{d}\,\mathrm{tr}\,X'(A + A')\,\mathrm{d}X = \mathrm{tr}(\mathrm{d}X)'(A + A')\,\mathrm{d}X.$$

Hence, by Proposition 18 (first identity) with $A \mapsto I_q$ and $B \mapsto A+A'$, the Hessian is $Hf(X) = I_q \otimes (A+A')$. □

Exercise 30 (Exercise 20 Cont'd)**.** Obtain the Hessian of $f(X) = \log|X'X|$, where X has full column rank.

Solution. Since $\mathrm{d}f(X) = 2\operatorname{tr} C'\,\mathrm{d}X$, where $C = X(X'X)^{-1}$, the second differential is

$$\begin{aligned}
\mathrm{d}^2 f(X) &= 2\,d\big(\operatorname{tr}(X'X)^{-1}X'\,\mathrm{d}X\big) \\
&= 2\operatorname{tr}(d(X'X)^{-1})X'\,\mathrm{d}X + 2\operatorname{tr}(X'X)^{-1}(\mathrm{d}X)'\,\mathrm{d}X \\
&= -2\operatorname{tr}(X'X)^{-1}(\mathrm{d}X'X)(X'X)^{-1}X'\,\mathrm{d}X + 2\operatorname{tr}(X'X)^{-1}(\mathrm{d}X)'\,\mathrm{d}X \\
&= -2\operatorname{tr}(X'X)^{-1}(\mathrm{d}X)'X(X'X)^{-1}X'\,\mathrm{d}X \\
&\quad - 2\operatorname{tr}(X'X)^{-1}X'(\mathrm{d}X)(X'X)^{-1}X'\,\mathrm{d}X + 2\operatorname{tr}(X'X)^{-1}(\mathrm{d}X)'\,\mathrm{d}X \\
&= 2\operatorname{tr}(X'X)^{-1}(\mathrm{d}X)'M\,\mathrm{d}X - 2\operatorname{tr} C'(\mathrm{d}X)C'\,\mathrm{d}X,
\end{aligned}$$

where $M = I_n - X(X'X)^{-1}X'$. The second equality in this derivation follows from considering $(X'X)^{-1}X'\,\mathrm{d}X$ as a product of three matrices: $(X'X)^{-1}$, X', and $\mathrm{d}X$ (a matrix of constants), the third equality uses the differential of the inverse in (34), and the fourth equality separates $\mathrm{d}X'X$ into $(\mathrm{d}X)'X + X'\,\mathrm{d}X$. Hence, we obtain from Proposition 18,

$$Hf(X) = 2(X'X)^{-1} \otimes M - 2K_{qn}(C \otimes C').$$

□

Exercise 31 (Exercise 21 Cont'd)**.** Let $f_k(X) = \operatorname{tr} X^k$ $(k = 1, 2, \dots)$. Find the Hessian.

Solution. We have $\mathrm{d}f_k(X) = k \operatorname{tr} X^{k-1}\,\mathrm{d}X$, and, in particular, $\mathrm{d}f_1(X) = \operatorname{tr}\mathrm{d}X$ and $\mathrm{d}f_2(X) = 2\operatorname{tr} X\,\mathrm{d}X$. Hence, $\mathrm{d}^2 f_1(X) = 0$ and, for $k \geqslant 2$,

$$\mathrm{d}^2 f_k(X) = k \operatorname{tr}(\mathrm{d}X^{k-1})\,\mathrm{d}X = k \sum_{j=0}^{k-2} \operatorname{tr} X^j (\mathrm{d}X) X^{k-2-j}\,\mathrm{d}X.$$

This gives $Hf_1(X) = 0$, and, for $k \geqslant 2$,

$$Hf_k(X) = (k/2) \sum_{j=0}^{k-2} K_n(X'^j \otimes X^{k-2-j} + X'^{k-2-j} \otimes X^j).$$

□

Exercise 32 (Exercise 22 Cont'd)**.** Find the Hessian of the matrix equation $F(X) = AX^{-1}B$, where X is nonsingular.

Solution. We know that $\mathrm{d}F(X) = -AX^{-1}(\mathrm{d}X)X^{-1}B$. The second differential is

$$\begin{aligned}\mathrm{d}^2 F(X) &= -A(\mathrm{d}X^{-1})(\mathrm{d}X)X^{-1}B - AX^{-1}(\mathrm{d}X)(\mathrm{d}X^{-1})B \\ &= 2AX^{-1}(\mathrm{d}X)X^{-1}(\mathrm{d}X)X^{-1}B.\end{aligned}$$

To obtain the Hessian matrix of the stth element of F, we let

$$C_{ts} = X^{-1} B e_t e_s' A X^{-1},$$

where e_s and e_t are elementary vectors with 1 in the sth (respectively, tth) position and zeros elsewhere. Then,

$$\mathrm{d}^2 F_{st}(X) = 2 e_s' A X^{-1} (\mathrm{d}X) X^{-1} (\mathrm{d}X) X^{-1} B e_t = 2 \operatorname{tr} C_{ts} (\mathrm{d}X) X^{-1} (\mathrm{d}X),$$

and hence

$$HF_{st}(X) = \frac{\partial^2 F_{st}}{(\partial \operatorname{vec} X)(\partial \operatorname{vec} X)'} = K_n (C_{ts}' \otimes X^{-1} + X'^{-1} \otimes C_{ts}).$$

□

12 Example 2: Maximum likelihood

$y_1, \ldots, y_n$ i.i.d. $N(\mu, \Omega)$, $\Omega \succ 0$, $n \geqslant m+1$. Loglikelihood:

$$\Lambda(\mu, \Omega) = -\frac{mn}{2} \log 2\pi - \frac{n}{2} \log |\Omega| - \frac{1}{2} \sum_{i=1}^{n} (y_i - \mu)' \Omega^{-1} (y_i - \mu). \tag{45}$$

Differentiating:

$$\begin{aligned}
\mathrm{d}\Lambda &= -\frac{n}{2}\,\mathrm{d}\log|\Omega| + \frac{1}{2}\sum_{i=1}^{n}(\mathrm{d}\mu)'\Omega^{-1}(y_i-\mu) \\
&\quad - \frac{1}{2}\sum_{i=1}^{n}(y_i-\mu)'(\mathrm{d}\Omega^{-1})(y_i-\mu) + \frac{1}{2}\sum_{i=1}^{n}(y_i-\mu)'\Omega^{-1}\,\mathrm{d}\mu \\
&= -\frac{n}{2}\,\mathrm{d}\log|\Omega| - \frac{1}{2}\sum_{i=1}^{n}(y_i-\mu)'(\mathrm{d}\Omega^{-1})(y_i-\mu) + \sum_{i=1}^{n}(y_i-\mu)'\Omega^{-1}\,\mathrm{d}\mu. \qquad (46)
\end{aligned}$$

Using $\mathrm{d}\log|\Omega| = \mathrm{tr}(\Omega^{-1}\,\mathrm{d}\Omega)$ and letting

$$S(\mu) = \frac{1}{n}\sum_{i=1}^{n}(y_i-\mu)(y_i-\mu)',$$

$$\mathrm{d}\Lambda = -\frac{n}{2}\,\mathrm{tr}(\Omega^{-1}\,\mathrm{d}\Omega + S\,\mathrm{d}\Omega^{-1}) + \sum_{i=1}^{n}(y_i-\mu)'\Omega^{-1}\,\mathrm{d}\mu. \qquad (47)$$

Setting $\mathrm{d}\Lambda = 0$ ($\hat{S} = S(\hat{\mu})$):

$$\mathrm{tr}\big(\hat{\Omega}^{-1} - \hat{\Omega}^{-1}\hat{S}\hat{\Omega}^{-1}\big)\,\mathrm{d}\Omega = 0, \qquad \sum_{i=1}^{n}(y_i-\hat{\mu})'\hat{\Omega}^{-1}\,\mathrm{d}\mu = 0,$$

for all $\mathrm{d}\Omega$ and all $\mathrm{d}\mu$. This, in turn, implies that

$$\hat{\Omega}^{-1} = \hat{\Omega}^{-1}\hat{S}\hat{\Omega}^{-1}, \qquad \sum_{i=1}^{n}(y_i - \hat{\mu}) = 0.$$

ML estimators:

$$\hat{\mu} = \bar{y}, \qquad \hat{\Omega} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})(y_i - \bar{y})'. \tag{48}$$

(Symmetry of Ω was not imposed, but $\hat{\Omega}$ is symmetric anyway.)
Second differential of (47):

$$\begin{aligned} \mathrm{d}^2\Lambda = &-\frac{n}{2}\operatorname{tr}\big((\mathrm{d}\Omega^{-1})\,\mathrm{d}\Omega + (\mathrm{d}S)\,\mathrm{d}\Omega^{-1} + S\,\mathrm{d}^2\Omega^{-1}\big) - n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu \\ &+ \sum_{i=1}^{n}(y_i - \mu)'(\mathrm{d}\Omega^{-1})\,\mathrm{d}\mu. \end{aligned} \tag{49}$$

For the information matrix, take expectations directly ($\mathsf{E}\, S = \Omega$, $\mathsf{E}\, \mathrm{d}S = 0$):

$$\begin{aligned}
\mathsf{E}\, \mathrm{d}^2\Lambda &= -\frac{n}{2}\operatorname{tr}\big((\mathrm{d}\Omega^{-1})\,\mathrm{d}\Omega + \Omega\,\mathrm{d}^2\Omega^{-1}\big) - n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu \\
&= \frac{n}{2}\operatorname{tr}\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}\,\mathrm{d}\Omega - n\operatorname{tr}(\mathrm{d}\Omega)\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1} - n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu \\
&= -\frac{n}{2}\operatorname{tr}\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}\,\mathrm{d}\Omega - n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu, \qquad (50)
\end{aligned}$$

using $\mathrm{d}\Omega^{-1} = -\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}$ and

$$\mathrm{d}^2\Omega^{-1} = 2\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}.$$

Symmetry of Ω brings in D_n. Vectorizing (50):

$$\begin{aligned}
-\mathsf{E}\, \mathrm{d}^2\Lambda &= \frac{n}{2}\operatorname{tr}\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}\,\mathrm{d}\Omega + n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu \\
&= \frac{n}{2}(\mathrm{d}\operatorname{vec}\Omega)'(\Omega^{-1}\otimes\Omega^{-1})\,\mathrm{d}\operatorname{vec}\Omega + n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu \\
&= \frac{n}{2}(\mathrm{d}\operatorname{vech}(\Omega))'D_m'(\Omega^{-1}\otimes\Omega^{-1})D_m\,\mathrm{d}\operatorname{vech}(\Omega) + n(\mathrm{d}\mu)'\Omega^{-1}\,\mathrm{d}\mu,
\end{aligned}$$

using Prop. 18 and $\mathrm{d}\operatorname{vec}\Omega = D_m\,\mathrm{d}\operatorname{vech}(\Omega)$. Information matrix for $(\mu, \operatorname{vech}\Omega)$:

$$\mathcal{I} = n\begin{pmatrix} \Omega^{-1} & 0 \\ 0 & \frac{1}{2}D_m'(\Omega^{-1}\otimes\Omega^{-1})D_m \end{pmatrix}. \qquad (51)$$

By Prop. 10:

$$(\mathcal{I}/n)^{-1} = \begin{pmatrix} \Omega & 0 \\ 0 & 2(D_m'D_m)^{-1}D_m'(\Omega \otimes \Omega)D_m(D_m'D_m)^{-1} \end{pmatrix}$$

and the determinant:

$$|(\mathcal{I}/n)^{-1}| = |\Omega| \cdot |2(D_m'D_m)^{-1}D_m'(\Omega \otimes \Omega)D_m(D_m'D_m)^{-1}| = 2^m|\Omega|^{m+2}.$$

13 Example 3: ML with parameters in the design matrix

Linear model $y = X\beta + u$, $\operatorname{var}(u) = \Omega(\theta)$, with $X = X(\psi)$. Goal: ML for (β, ψ, θ) and the information matrix.

Normality loglikelihood:

$$\Lambda = \text{const.} - \tfrac{1}{2}\log|\Omega| - \tfrac{1}{2}(y - X\beta)'\Omega^{-1}(y - X\beta). \tag{52}$$

With $Z = \partial \operatorname{vec} X/\partial\psi'$ and Prop. 6:

$$\mathrm{d}(X\beta) = X\,\mathrm{d}\beta + (\beta' \otimes I_n)Z\,\mathrm{d}\psi.$$

Differentiating Λ:

$$\begin{aligned} \mathrm{d}\Lambda = &-(1/2)\operatorname{tr}(\Omega^{-1}\,\mathrm{d}\Omega) + (1/2)(y - X\beta)'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}(y - X\beta) \\ &+ (y - X\beta)'\Omega^{-1}X\,\mathrm{d}\beta + (y - X\beta)'\Omega^{-1}(\mathrm{d}X)\beta, \end{aligned} \tag{53}$$

FOCs for β, θ, ψ:

$$
\begin{aligned}
(y - X\beta)'\Omega^{-1} X \, \mathrm{d}\beta &= 0, \\
(y - X\beta)'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}(y - X\beta) &= \operatorname{tr}(\Omega^{-1} \, \mathrm{d}\Omega), \\
(y - X\beta)'\Omega^{-1}(\mathrm{d}X)\beta &= 0.
\end{aligned}
$$

The first gives

$$
\hat{\beta}(\psi, \theta) = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y. \tag{54}
$$

Concentrating out β:

$$
\Lambda_c(\psi, \theta) = \text{const.} - \tfrac{1}{2}\log|\Omega| - \tfrac{1}{2}\hat{u}'\Omega^{-1}\hat{u}, \quad \hat{u} = y - X\hat{\beta}.
$$

Grid-search ψ: for fixed $\psi = \psi_0$ ($\mathrm{d}\psi = 0$),

$$
\begin{aligned}
d\hat{\beta} &= [d(X'\Omega^{-1}X)^{-1}]X'\Omega^{-1}y + (X'\Omega^{-1}X)^{-1} \, \mathrm{d}(X'\Omega^{-1}y) \\
&= -(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}\hat{u},
\end{aligned}
$$

and hence

$$
\begin{aligned}
\mathrm{d}\Lambda_c(\psi_0, \theta) = &-(1/2)\operatorname{tr}(\Omega^{-1} \, \mathrm{d}\Omega) + (1/2)\hat{u}'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}\hat{u} \\
&- \hat{u}'\Omega^{-1}X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}\hat{u}.
\end{aligned}
$$

Solve $\mathrm{d}\Lambda_c = 0$ for $\hat{\theta}(\psi_0)$, maximize $\Lambda_c(\psi, \hat{\theta}(\psi))$ over ψ, then recover $\hat{\beta}$.

Information matrix. Let $\mu = X\beta$. Differentiating (53):

$$\begin{aligned} \mathrm{d}^2\Lambda = {} & (1/2)\operatorname{tr}(\Omega^{-1}\,\mathrm{d}\Omega)^2 - (y - X\beta)'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}(y - X\beta) \\ & - (\mathrm{d}\mu)'\Omega^{-1}(\mathrm{d}\mu) - 2(y - X\beta)'\Omega^{-1}(\mathrm{d}\Omega)\Omega^{-1}(\mathrm{d}\mu) + (y - X\beta)'\Omega^{-1}(d^2\mu) \\ & - (1/2)\operatorname{tr}(\Omega^{-1}\,\mathrm{d}^2\Omega) + (1/2)(y - X\beta)'\Omega^{-1}(\mathrm{d}^2\Omega)\Omega^{-1}(y - X\beta). \end{aligned} \tag{55}$$

Taking expectations:

$$-\mathsf{E}(\mathrm{d}^2\Lambda) = \tfrac{1}{2}\operatorname{tr}(\Omega^{-1}\,\mathrm{d}\Omega)^2 + (\mathrm{d}\mu)'\Omega^{-1}(\mathrm{d}\mu).$$

So $\mathcal{I}$ is block-diagonal in (β, ψ) vs. θ. Writing

$$\operatorname{tr}(\Omega^{-1}\,\mathrm{d}^2\Omega) = (\mathrm{d}\operatorname{vec}\Omega)'(\Omega^{-1} \otimes \Omega^{-1})(\mathrm{d}\operatorname{vec}\Omega) = (\mathrm{d}\theta)'\mathcal{I}_{\theta\theta}\,\mathrm{d}\theta,$$

with

$$\mathcal{I}_{\theta\theta} = \left(\frac{\partial \operatorname{vec}\Omega}{\partial\theta'}\right)'(\Omega^{-1} \otimes \Omega^{-1})\left(\frac{\partial \operatorname{vec}\Omega}{\partial\theta'}\right),$$

and

$$(\mathrm{d}\mu)'\Omega^{-1}(\mathrm{d}\mu) = \begin{pmatrix}\mathrm{d}\beta \\ \mathrm{d}\psi\end{pmatrix}' \begin{pmatrix}\mathcal{I}_{\beta\beta} & \mathcal{I}_{\beta\psi} \\ \mathcal{I}_{\psi\beta} & \mathcal{I}_{\psi\psi}\end{pmatrix} \begin{pmatrix}\mathrm{d}\beta \\ \mathrm{d}\psi\end{pmatrix},$$

with

$$\mathcal{I}_{\beta\beta} = X'\Omega^{-1}X, \qquad \mathcal{I}_{\beta\psi} = \mathcal{I}'_{\psi\beta} = X'(\beta' \otimes \Omega^{-1})Z,$$

and

$$\mathcal{I}_{\psi\psi} = Z'(\beta\beta' \otimes \Omega^{-1})Z,$$

the information matrix is

$$\mathcal{I} = \begin{pmatrix} \mathcal{I}_{\beta\beta} & \mathcal{I}_{\beta\psi} & 0 \\ \mathcal{I}_{\psi\beta} & \mathcal{I}_{\psi\psi} & 0 \\ 0 & 0 & \mathcal{I}_{\theta\theta} \end{pmatrix}.$$

$\mathcal{I}^{-1}$ approximates var(MLE).

14 Example 4: The Eckart–Young theorem

Best rank-r approximation of A $(m \times n)$ by XZ' with X $(m \times r)$, Z $(n \times r)$, $Z'Z = I_r$. Minimize

$$f(X, Z) = \operatorname{tr}(A - XZ')(A - XZ')'. \tag{56}$$

Theorem 3 (Eckart–Young)**.** *The minimum of f is obtained when Z contains the eigenvectors associated with the r largest eigenvalues of $A'A$ and $X = AZ$. The best approximation $\tilde{A}$ (of rank r) to A is then $\tilde{A} = AZZ'$, and the constrained minimum of f is the sum of the $n - r$ smallest eigenvalues of $A'A$.*

Proof. Lagrangian with symmetric L (Exercise 18):

$$\mathcal{L}(X, Z) = \tfrac{1}{2}\operatorname{tr}(A - XZ')(A - XZ')' - \tfrac{1}{2}\operatorname{tr} L(Z'Z - I_r). \tag{57}$$

Using $L = L'$ and trace cyclicity,

$$\mathrm{d}\mathcal{L} = -\operatorname{tr}(A - XZ')Z(\mathrm{d}X)' - \operatorname{tr}(X'A - X'XZ' + LZ')\,\mathrm{d}Z.$$

FOCs:

$$\begin{aligned} (A - XZ')Z &= 0, \\ X'A - X'XZ' + LZ' &= 0, \\ Z'Z &= I_r. \end{aligned} \tag{58}$$

From the first FOC and $Z'Z = I_r$: $X = AZ$. Substituting into the second and postmultiplying by Z gives $L = 0$, so

$$(A'A)Z = Z(Z'A'AZ). \tag{59}$$

Diagonalize $Z'A'AZ = P\Lambda_1 P'$ (orthogonal P, diagonal Λ_1), set $T_1 = ZP$. Then $A'AT_1 = T_1\Lambda_1$ with $T_1'T_1 = I_r$: T_1 holds r eigenvectors of $A'A$, Λ_1 the matching eigenvalues.
Since $I - ZZ'$ is idempotent,

$$\operatorname{tr}(A - XZ')(A - XZ')' = \operatorname{tr} A(I - ZZ')A' = \operatorname{tr} A'A - \operatorname{tr}\Lambda_1.$$

$$\operatorname{tr}(A - XZ')(A - XZ')' = \operatorname{tr} A'A - \operatorname{tr}\Lambda_1. \tag{60}$$

Minimize $\iff$ maximize $\operatorname{tr}\Lambda_1$: take the r largest eigenvalues. Best approximation $\tilde{A} = AT_1T_1'$; min value = sum of $n - r$ smallest eigenvalues of $A'A$. $\square$